

Model Evaluation in DL

=> Model yaxshi o'rganayotgandek tuyuladi lekin real hayotda yaxshi natija bermaydi

=> Quyidagi diagnose larni o'z ichiga k- k.

1. Model yaxshi o'rganayabdimi?
2. Overfitting emismi?
3. Generalization bo'ldimi?
4. Modelni qanday rivojlantiramiz?

Overfitting

=> Deep learningda model featurlarni avtomatik o'rganadi. Feature engineering ni o'zi qiladi. (SML da semiz qildik)

=> Deep learningda ko'proq data ko'k.

=> ko'p data bo'lgandan so'ng para- metrlar ham tabiiy ko'p bo'ldi

=> Natijada Overfitting ehtimoli oshadi.

=> Shu sababli DL da evaluation muhimroq SML ga qaraganda.

Loss function

=> Butun jarayon Loss ni kamay- tirilga qaratilgan.

=> Loss ni kamaytirilgan o'zi yetarli emas.

=> Lossni & qitmo!
Train loss va validation loss
orqali tekshiramiz. (Trainingda)

=> Final metrics orqali modelni
bahalaymiz.

NN

=> Neuronlar - kichik qatorlar, ya'ni
NN ning bir qitmi.

=> NN - katta qator. Neuronlarning
jami soniga ts

=> Activation - Neural network non-linearity
g'hadir. Agar g'hadir (SAL
dan fargi bo'lmay qoladi.

=> Bu sababli bu holatda ham
overfittinga moyil bo'lib qoladi.

Evaluation - Qanday muammo borligini
keltiradi

Bias-variance - Nima uchunligini.

Tuning - Nima o'zgartirilishi
keltiradi.

Evaluation Process

- Data ni 3 qismga bōlami².
- Train dataset - modelni train qilish uchun.
- Validation dataset - train jarayonida tekshirish uchun.
- Test dataset - Modelni oxirida evaluate qilish uchun.

Loss function $\Rightarrow$ Model training jarayoni da ishlatilinao²li. (train set)

Evaluation metric $\Rightarrow$ Modelni evaluate qilish uchun. (validation va test dataset).

High bias $\Rightarrow$ underfitting

Low bias $\Rightarrow$ Overfitting

High variance $\Rightarrow$ Overfitting

Low variance $\Rightarrow$ underfitting

High bias (underfitting) - kattaroq model dan foydalanib, trainni uzaytirib, regularizationni qo'llab fik qilamiz.

High variance (overfitting) - koproq datadan foydalanib, dropout, L2, early stopping (regularization techniques) va kichikroq model orqali fik qilamiz.

Hyperparameter tuning

learning rate, hidden layers size, number of layers, batch size, optimizer, dropout, weight decay and number of epochs larvi ispartirib best topikni jarayoni.

SML va DPL da hyperparameter tuning metodlari deyiladi bir xil.

Grid Search ni DPL ishlatmaymiz.

chunki: training time juda ko'p olib ketadi.

Eng ko'p ishlatiladigan:

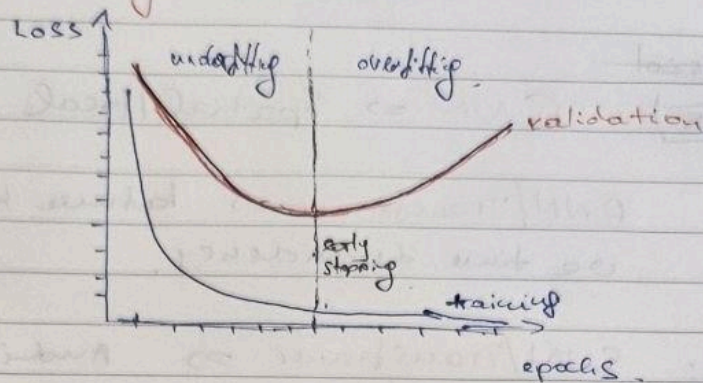
Random Search

Manual Tuning

Learning rate Scheduling

Population based tuning.

Learning curves.



1. pastlap loss trainingda ham validationda ham juda yuqori edi.

train $\Rightarrow$ high
val $\Rightarrow$ high } underfitting

2. Dastlab train zilil loss lovu kamaytirishga eriklayotgan edi (epochs soni ortitilni train zilil lovu). Lekin validation loss eng low zignatiga ega boldi va yana ortitilni boshladi.

Ya'ni: train $\Rightarrow$ low.

val $\Rightarrow$ high bolishni boshladi.

Bu jarayon davom etganda Overfitting ga olib kelardi.

Lekin: early stopping ni qo'llab
buni oldini olish.

Neural networks

~~Structured~~

Images: CNN $\Rightarrow$ Spatial/Local patterns.

Text: RNN/Transformer $\Rightarrow$ ketura ketik
or time dependency.

Audio: CNN/Transformer $\Rightarrow$ Audio signal-
ing temporal/spectral structure.

Time series: RNN/LSTM/Transformer $\Rightarrow$
(Recurrent NN).

CNN $\Rightarrow$ Convolutional NN.

RNN $\Rightarrow$ Recurrent NN.

LSTM $\Rightarrow$ Long short term memory.

MLP $\Rightarrow$ Multi-layer perceptron.

Structured/Tabular: Fully connected network (MLP).

!!
Structured NN

Tabular NN $\Rightarrow$ Neural network for tabular data

Classifier NN $\Rightarrow$ Neural network classifier.

Fraud/detection NN $\Rightarrow$ Specific application.

Health Prediction NN $\Rightarrow$ Domain specific.